

Your Name

111-111-1111 | [outlook.com](#) | [LinkedIn](#) | [Github](#) | [Portfolio](#)

EDUCATION

University

Canada

Software Engineering (BS)

Expected Graduation: Dec 2026

- **Related Coursework:** Applied Machine Learning, Data Structures & Algorithms, Java OOP, Python Computing, Software Quality Assurance, Databases I, Software Construction, Real-Time Embedded Software

EXPERIENCE

Telecommunications x2

Canada

Full Stack Machine Learning Intern

May 2025 - Dec 2025

- Built an **OODA-based autonomous app** with React, PostgreSQL and FastAPI (Python) to manage 5G Core workloads—addressing UPF congestion, auto-scaling, and **cutting deployment time by 80%**.
- Designed a cloud-native **multi-agent LLM system** using the A2A protocol, enabling **RAG-driven chain-of-thought** analysis and network reconfiguration via MCP servers to autonomously resolve 5G network issues.
- Built and benchmarked CPU utilization prediction models (Prophet, ARIMA, LSTM) in **PyTorch/scikit-learn**, selecting the best model to reduce MAE from 2.6 → 1.6 (-38%) on production metrics.
- Built **Go Kubernetes Operators** that converted ML workflow requests into automated training/deployment jobs, integrating **Kafka** for event-driven execution and **Crossplane** for infrastructure provisioning.

Government of Canada

Canada

Software Developer Intern

May 2024 - Dec 2024

- Built **React/Django** dashboards and on-demand exports for the FSEIS application, cutting report generation time from 10–15 minutes to less than 1 minute for LabsCanada users.
- Migrated cloud infrastructure from **Azure to AWS**, remapping services and configurations and using AWS DMS for data migration to support improved scalability and operations.
- Implemented **RTL/Jest UI tests and PyTest API tests** with service mocking and CI coverage gates, cutting manual regression testing 50% and reducing release risk.

University

Canada

Full Stack Developer Intern

Sep 2023 - Mar 2024

- Engineered a reactive **Angular** registration flow for OutStem with multi-step forms, autosave, and inline validation, reducing user friction and support-driven rework.
- Built a **Python service using OpenAI** to convert free-text user intent into validated structured filters (JSON) for 500+ client attributes, enabling dynamic filter suggestions and reducing “no results” searches.
- Optimized PostgreSQL and Java REST APIs by removing N+1 patterns and rewriting slow queries, cutting query latency 50% and reducing peak-hour timeouts.

Telecommunications

Canada

Software Developer Intern

Jan 2023 - Apr 2023

- Extended CKEY, an open-source **IAM**, in **Java/Quarkus** to support SSO and auth flows for internal RESTful services, web apps, and mobile clients.
- Optimized Kubernetes CPU/memory requests and limits and validated horizontal/vertical scaling via load testing, **increasing throughput by 40%**.
- Reduced manual test scripting 80% by automating 15+ admin/auth flows via Keycloak Benchmark integration in CKEY's Quarkus test suite, increasing endpoint coverage by 35%.

PROJECTS

Keyrex Capstone — Github | Next.js, React, AWS, PostgreSQL, Drizzle ORM, SST

- Built a full-stack key management system for the University, enabling digital key requests/approvals, role-based access, and audit-ready tracking of key custody.

TECHNICAL SKILLS

Languages: Python, JavaScript, TypeScript, Java, Go, C, SQL

Frameworks/Libraries/Misc: React, Angular, Django, FastAPI, Flask, Next.js, PostgreSQL, Spring Boot, Node.js, Express.js, MongoDB, Prometheus, Cypress, Tailwind

Dev/Cloud Tools: AWS (EC2, S3, Lambda), Microsoft Azure, Linux, Kubernetes, Docker, Git, Helm, Kafka, Jenkins